

Effectiveness of honey bees for applying the biocontrol agent *Gliocladium roseum* to strawberry flowers to suppress *Botrytis cinerea*

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An inoculum dispenser designed to fit in bee hives was developed to contaminate honey bees with *Gliocladium roseum*. In the greenhouse, bees emerging from a dispenser containing corn meal infested with 10^7 cfu *G. roseum*/g each acquired $3.6\text{--}32.0 \times 10^4$ cfu *G. roseum*; in the field, bees emerging from dispensers containing talc-corn meal infested with 5×10^8 cfu/g each carried $8.8\text{--}180 \times 10^4$ cfu. Infested substrate was visible macroscopically and microscopically on almost all external surfaces of bees that had crawled in substrate. Numerous conidia were observed adhering to femoral setae and other parts of bees legs examined on a scanning electron microscope. Inoculum density of *G. roseum* on strawberry flowers visited by contaminated bees averaged 22 000 and 1500 cfu/flower, respectively, in two greenhouse studies and ranged from 1600 to 27 000 and from 300 to 15 000 cfu/flower, respectively, in two field studies. Inoculum transfer was greater when conditions favored foraging of the bees on the flowers. In the field, inoculum density of *G. roseum* on flowers inoculated by bees usually was as high as or higher, and more stable, than on flowers of plants sprayed weekly with a spore suspension of 10^7 conidia *G. roseum*/mL water plus surfactant. In field plots (but not in field microplots) *G. roseum* significantly suppressed incidence of *B. cinerea* on petals, stamens, and fruit when applied as sprays or by bees, and effectiveness of the application methods did not, on average, differ significantly. It is concluded that bees effectively transferred inoculum of *G. roseum* from the dispensers to strawberry flowers, and that the transferred inoculum usually suppressed *B. cinerea* on the flowers and fruits.

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On a fabriqué un distributeur d'inoculum de *Gliocladium roseum* conçu pour aller à l'intérieur des ruches. A la sortie du distributeur qui contenait de la farine de maïs avec 10^7 ufc de *G. roseum*/g, chaque abeille portait de $3,6$ à $32,0 \times 10^4$ ufc de *G. roseum* lors d'expériences en serres. Au champ, et avec un distributeur contenant de la farine de maïs et du talc infestée de 5×10^8 ufc/g, chaque abeille portait $8,8\text{--}180 \times 10^4$ ufc. Le substrat infesté était visible de façon microscopique et macroscopique sur à peu près toutes les surfaces externes des abeilles qui avaient passé dans le substrat. On a observé de nombreuses conidies qui adhéraient aux poils fémoraux et aux autres parties des pattes des abeilles lors d'un examen au microscope électronique à balayage. La densité d'inoculum de *G. roseum* sur les fleurs de fraisiers butinées par les abeilles infestées fut en moyenne de 22 000 et 1500 ufc/fleur, respectivement, lors des deux études en serres, et a varié de 1600 à 27 000 et de 300 à 15 000 ufc/fleur, respectivement, lors des deux expériences au champ. Le transfert d'inoculum fut plus élevé quand les conditions favorisaient la recherche de nectar sur les fleurs. Au champ, la densité d'inoculum de *G. roseum* sur les fleurs inoculées par les abeilles fut habituellement aussi élevée ou plus élevée, et plus stable, que sur les fleurs des plantes auxquelles on avait appliqué à chaque semaine une suspension de spores de 10^7 conidies de *G. roseum*/mL d'eau avec de l'adhérent. Dans les parcelles au champ (mais non les micro-parcelles), le *G. roseum* a significativement réduit l'incidence du *B. cinerea* sur les pétales, les étamines et les fruits, autant lorsqu'on l'a répandu sous forme de pulvérisations ou à l'aide des abeilles. En moyenne, l'efficacité de ces deux méthodes d'application ne fut pas significativement différente. On conclut que les abeilles transfèrent efficacement l'inoculum de *G. roseum* des distributeurs aux fleurs de fraisiers et que l'inoculum transféré a habituellement réprimé le *B. cinerea* sur les fleurs et les fruits.

Gliocladium roseum Link: Bainier is an effective biocontrol agent for gray mold of strawberry (*Fragaria* $\times$ *ananassa* Duch.) caused by *Botrytis cinerea* Pers.: Fr. (Peng & Sutton 1990a, 1990b, 1991a). The fungus controlled gray mold fruit rot when applied as conidial suspensions to strawberry

at the flowering and fruiting stages at least as effectively as the fungicide captan (Peng & Sutton 1991a). The chief target of conventional fungicides and experimental biocontrol agents used to control gray mold fruit rot is the flowers, which normally are infected by *B. cinerea* as a prelude to invasion

of the fruits (Powelson 1960, Jarvis 1962, Jarvis & Borecka 1968, Bristow et al. 1986). However, spray applications are inefficient because only a small proportion of applied material is deposited on the flowers, and because flowers that open after spraying may remain unprotected (Jarvis 1962, Sutton 1990). Development of an efficient method for applying the biocontrol agent to strawberry flowers may improve the feasibility and effectiveness of biocontrol on the farm, should *G. roseum* or other agent be registered for use against gray mold fruit rot (Sutton 1991).

The need to apply *G. roseum* specifically to the flowers of strawberry prompted consideration of the European honey bee (*Apis mellifera* L.) as a possible delivery vehicle. Bees are well-known vectors not only of pollen, but also of fungi and bacteria among flowers of many plant species (Free 1970, Batra et al. 1973, Harrison et al. 1980, Sandhu & Waraich 1985, Thomson et al. 1990). Although not strongly attracted to strawberry flowers, bees may visit frequently when more attractive flowers are lacking or when the bees are sufficiently numerous to remove nectar and pollen from competing sources (Levin 1978, Waser 1983, Ramsay 1987, Bagnara & Vincent 1988, Larsson et al. 1991). Chemical attractants also may be used to increase visits (Free 1970, McGregor 1976). Visits of honey bees to strawberry flowers are relatively frequent in warm sunny weather but infrequent or lacking when conditions are cool or rainy (Free 1968a, 1968b). Use of honey bees is recommended in Ontario and elsewhere for optimal pollination of strawberries (Free 1970, McGregor 1976, Lockett 1985, Antonelli et al. 1988, Evans et al. 1988, Kevan 1988).

The purpose of the present study was to examine European honey bees as a potential means for applying *G. roseum* to strawberry flowers. The study necessitated evaluation of substrates for inoculum production of the biocontrol agent, and development of an inoculum dispenser for contaminating the bees. Transmission of *G. roseum* to strawberry flowers was investigated in the greenhouse and in the field, and effectiveness of the bee-transmitted inoculum in suppressing gray mold was examined.

Materials and methods

Inoculum production. Isolate PG-A-Fr-87-001 of *B. cinerea*, obtained from a strawberry fruit at the Arkell Research Station near Guelph in 1987, was used for the studies. Conidia of the pathogen were produced on a strawberry agar medium (Braun & Sutton 1988) and recovered by aspiration into sterile distilled water plus surfactant (0.05% Triton XR-100). The concentration of suspended conidia was

estimated with the aid of a hemacytometer and adjusted to 2×10^3 spores/mL.

The biocontrol agent, *G. roseum* isolate PG-A-Fr-88-710, was obtained from a strawberry fruit at the Arkell Research Station in 1988 (Peng & Sutton 1991a). To produce conidia, the fungus was grown on potato dextrose agar (PDA, pH 7.0) in plastic petri dishes beneath cool-white fluorescent lamps (14-h photoperiod) at 20–23°C for 15–20 d. For recovering conidia, the colonies were flooded with sterile distilled water plus the surfactant and rubbed with a bent glass rod. Conidial suspensions were filtered using two layers of cheese cloth, and spore concentrations were estimated using a hemacytometer.

Various substrates were evaluated for producing inoculum of *G. roseum* to potentially contaminate honey bees. Corn meal (300 g), poultry layer's mash (United Cooperatives of Ontario, Mississauga, ON L5A 3A4: 300 g), wheat bran (50 g), and oat bran (50 g) were placed in separate 1-L screw-capped jars (Pyrex®), moistened with 50 mL distilled water, autoclaved at 121°C and 103 kPa for 20 min, and cooled. After addition of 50 mL spore suspension (10^7 conidia/mL) of *G. roseum* each jar was shaken vigorously by hand. The jars, with caps loosened, were kept on the laboratory bench for 3 d after which the infested substrates were transferred to ethanol-sterilized plastic trays (70 × 50 × 5 cm) to form a layer about 5 mm deep in each tray. The trays were covered with transparent plastic film and kept beneath the fluorescent lamps at 20–23°C for 15 d. The film was subsequently removed and the substrates were allowed to air dry for 2 d. The infested substrates were pulverized in a domestic blender operated at high speed for 1 min and stored in the glass jars at about 4°C for 1–2 wk before use. In a second method of inoculum production, *G. roseum* was grown on PDA for 20 d and air dried for 2 d as described above. Sixty dried cultures were blended with 10 g talc and 1 g corn meal in a domestic blender operated at high speed for 1 min. The mixture was screened through 2 mm nylon mesh to remove flakes of dried PDA.

To estimate inoculum concentration of *G. roseum* in each infested substrate, six 1-g samples taken at random were each suspended in 250 mL sterile distilled water and agitated on a rotary shaker at 110 rpm for 1 h. Five aliquots (0.1 mL) of ten-fold serial dilutions of each suspension were spread on PDA amended with Triton XR-100 and streptomycin sulfate (final concentrations of 0.2% and 100 ppm, respectively) in petri dishes. The medium is referred to as PDTSA and was modified from that of VanEtten and Kolmark (1977). The dishes were kept on the laboratory bench at 20–23°C for 5–7 d,

colonies of *G. roseum* were counted and inoculum concentrations estimated. Long term viability of inoculum in the talc-corn meal formulation stored in jars at 4–5°C was examined by estimating inoculum density when the formulation was freshly prepared and at 4-wk intervals thereafter.

Bee colonies. Nuclear colonies of honey bees each with a queen and about 5000 workers in half-sized hives were used for inoculum acquisition and transmission studies in the laboratory, greenhouse, and field. Before use, the colonies were maintained at the University of Guelph Apiculture Laboratory. Colonies in the greenhouse were allowed to acclimate for 7 d before experiments began, and were supplied with honey within the hive and water outside the hive.

Inoculum dispenser. A dispenser for contaminating bees with inoculum of the biocontrol agent was developed from pollen dispensing devices (King & Burrell 1933, Townsend et al. 1958, Free 1970, Karmo & Vickery 1987). Constructed largely of wood, hardware cloth, and Plexiglas®, the dispenser was designed to fit in the lower portion and front of the half-sized hive (Fig. 1, A and B). Inoculum was spread about 5–8 mm deep on the Plexiglas® tray (15.5 cm × 15.5 cm). When the dispenser was positioned in the hive a corridor about 1 cm high was formed. Outgoing bees crawled about 12 cm through the inoculum in the corridor towards daylight transmitted through the front of the tray, and then upwards and through an exit slot. Incoming bees entered the hive below the tray.

Inoculum acquisition by bees. Inoculum acquisition was studied in petri dishes and in dispensers mounted in hives in the greenhouse and in the field. Inoculum preparations of *G. roseum* on corn meal and talc-corn meal, each containing 10^8 cfu/g, and noninfested control media, were placed in 9-cm-diam petri dishes (5 g formulation/dish). Four bees were placed in each of five replicate dishes of each formulation and allowed to crawl for 5 min. The bees were then killed with vaporized ethyl acetate. Two of the bees in each dish were placed in 100 mL sterile distilled water plus surfactant in a 250 mL erlenmeyer flask and shaken for 1 h on a rotary shaker operated at 110 rpm. Aliquots (0.1 mL) of serial dilutions of the aqueous suspension in each flask were spread onto PDTSA, incubated as before, and colonies of *G. roseum* were counted after 5–7 d. The remaining two bees in each dish were examined on a dissecting microscope. Bees to be observed with a scanning electron microscope (SEM) were treated with drops of a 3:97 mixture of glacial acetic acid and 95% ethanol to fix conidia of *G. roseum*, and air dried for 1 wk. Dried legs of the bees were mounted

on SEM stubs, coated with a thin layer of gold-palladium mixture using a sputter coater (model Hummer VII, Anatech, Alexandria, VA 22310) and viewed on an SEM (Hitachi 5570, Japan).

In both the greenhouse and field studies, twelve honey bees were captured at random as they emerged from the inoculum dispenser on the hive. The dispenser in the greenhouse contained inoculum on corn meal (10^7 cfu/g), but in the field the talc-corn meal formulation (5×10^8 cfu/g) was used. In both studies, the dispenser was replenished with inoculum in the evening before the day of capture of the bees. Inoculum of *G. roseum* on each of 12 bees per substrate was estimated by washing and dilution plating as described above.

Estimation of gray mold. Disease was quantified indirectly by estimating sporulation potential of *B. cinerea* in strawberry tissues (Peng & Sutton 1991a). Flowers in greenhouse studies were surface sterilized in 70% ethanol for 5 s and in 0.6% NaOCl for 60 s, rinsed twice in sterile distilled water, and the petals and stamens were detached and transferred to paraquat-chloramphenicol agar medium (PCA; Bannon 1978, Peng & Sutton 1991a) in petri dishes. Sporulation incidence on stamens and sporulation density on petals were estimated using a dissecting microscope after 7 d. Sporulation density was estimated from direct counts of conidiophores and areas of petals measured with a digitizer (model MOP-3, Zeiss Inc.). Petals and stamens of flowers in field studies were transferred directly to PCA without surface sterilization. Plated tissues were incubated beneath cool-white fluorescent tubes (14-h photoperiod) at 20–23°C for 7 d, then examined as before for sporulation incidence of *B. cinerea* (Peng & Sutton 1991a). Ripe strawberry fruits harvested from field plots were placed separately into compartments (4 cm × 4 cm × 4.5 cm deep) of plastic plant flats, which were stacked in a cold room (5–8°C). The fruits were examined after 15 d for signs of *B. cinerea*.

Transfer of *G. roseum* in the greenhouse. Plants of strawberry cv. Chandler were grown to the flowering stage in a greenhouse that was free from other plant materials. The plants were maintained in 12-cm pots containing artificial soil mix (Promix®, Plant Products Ltd., Brampton, ON, L6T 1G1) and supplied at 2-wk intervals with soluble 10-52-10 fertilizer (total N, P₂O₅, and K₂O, respectively) at a concentration of 5 g/L water. The plants were assigned randomly to groups, each bearing 20–25 flowers.

Vectoring studies were conducted on sunny days inside a tent (4 m × 2 m × 2 m) constructed in the greenhouse from translucent screening (2 mm nylon

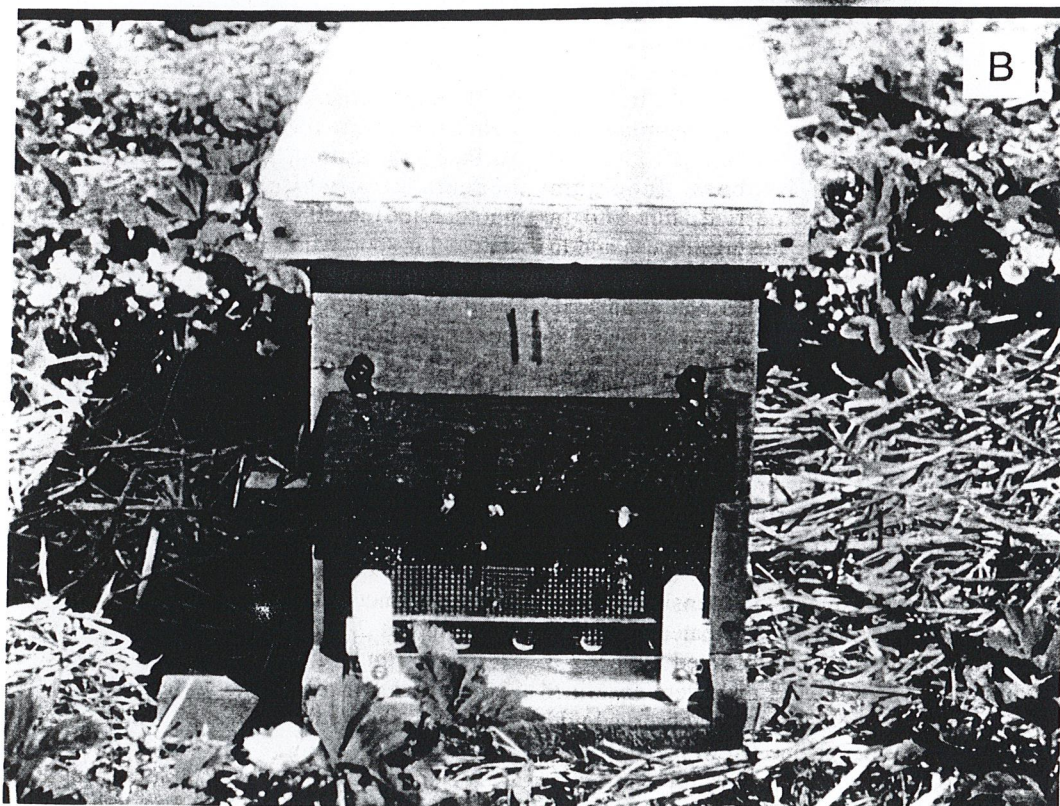
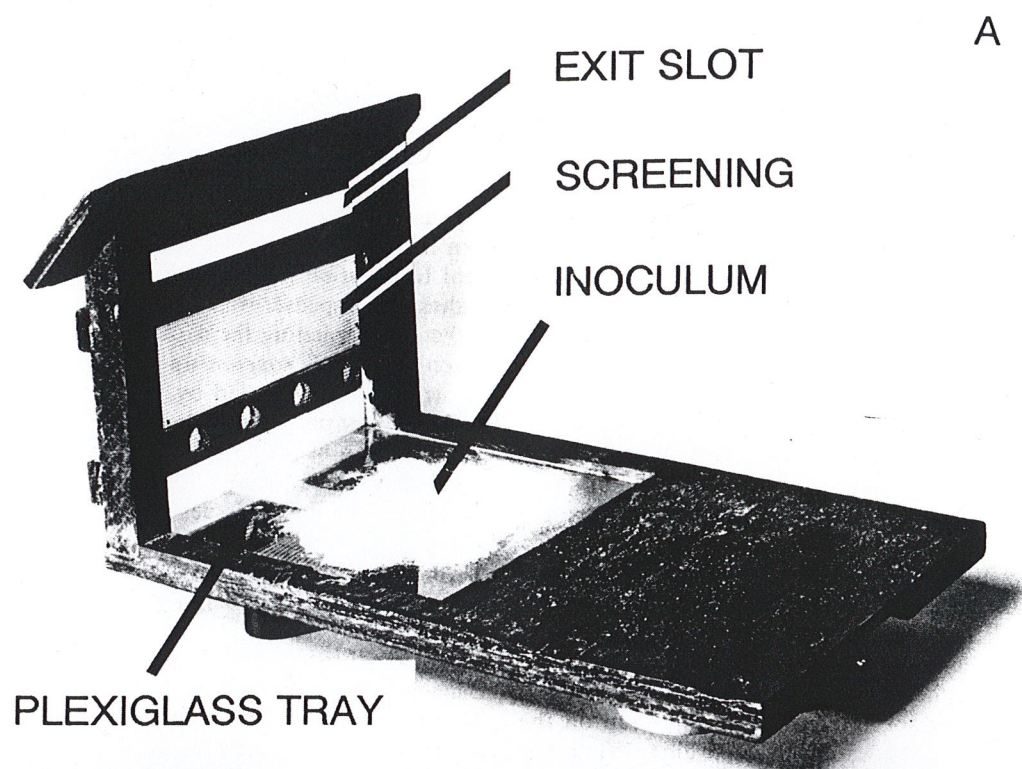


Figure 1. An inoculum dispenser shown A) separately and B) attached to a half-sized hive in a strawberry plot.

mesh). A half-sized hive of bees with an inoculum dispenser was placed inside the tent. In a control treatment, noninfested cornmeal was placed in the dispenser, and the bees were allowed to forage on six replicate groups of the flowering strawberries positioned randomly inside the tent. Foraging was permitted for a 24-h period initiated at 0900 hours. The noninfested substrate was replaced in the dispenser with corn meal infested with *G. roseum* (10^7 cfu/g), and the bees were allowed to forage for 24 h, as before, on a second set of six groups of flowering strawberry plants.

Immediately following the foraging periods, two flowers of each replicate group of plants were placed in 100 mL sterile distilled water plus 50 μ L Triton X-100 in a 250 mL erlenmeyer flask and shaken on a rotary shaker at 110 rpm. After 60 min, three 0.1 mL aliquots of the wash water were each spread on the surface of PDTSA in a petri dish. The dishes were incubated and colonies of *G. roseum* were counted as described above. All other flowers of each treatment were inoculated to incipient runoff with a spore suspension (2×10^3 conidia/mL) of *B. cinerea* using an aerosol sprayer when the second group of plants was removed from the tent. Inoculated plants were kept in a mist chamber for 36 h, after which gray mold was estimated on 20 petals and 120 stamens of plants of each replicate group in each treatment.

Air temperature and relative humidity (r.h.) were measured continuously near the plants in the greenhouse with a calibrated hygrothermograph (Lambrecht model 252, Göttingen, Germany). The experiment was repeated once.

Inoculation studies in the field. Effectiveness of inundative sprays and honey bee vectors as means for applying inoculum of *G. roseum* to strawberry flowers were compared in a matted-row planting (14×30 m) of cv. Redcoat at the Arkell Research Station in 1990. The planting was 3 years old and managed according to Ontario recommendations (Evans et al. 1988) except that no pesticides or overhead irrigation were applied during the crop season. The strawberry rows were each 0.5 m wide and spaced on centers 1 m apart. Treatment plots each consisted of 4 m of row and were spaced 1 m apart within the row. Spray-treated plots and untreated check plots were covered throughout flowering with black fiberglass screen of 2 mm mesh to prevent visits by bees. The screen was suspended 20–40 cm above the foliage on wire arches. For spray treatments, the covers were temporarily removed and a spore suspension of *G. roseum* (10^7 conidia/mL water plus surfactant) was applied to form a fine wetness deposit primarily on the flowers, using a compressed-air sprayer of 7.6 L capacity and

equipped with a single nozzle. Inoculum was applied during 1 to 2 h before dusk on 27 May and 1 and 7 June, when the phenological stage of the strawberries ranged from about 20% to 85% of flowers opened. For the bee-vectoring treatment, three half-sized hives, each with a dispenser, were positioned among the plots on 24 May. Talc-corn meal formulation of *G. roseum* (5×10^8 cfu/g) was placed in the dispensers after nightfall on 27 May and at intervals of 1–2 d until 14 June. All plants in the plots and surrounding area of the planting were sprayed with a bee-attractant (Bee-Scent®, Scentry Inc., Buckeye, AZ 85326) at 2–5 d intervals or immediately after rain. Plots covers were removed temporarily to allow application of the attractant, and permanently on 16 June, when flowering was almost completed and the fruits were green to white. Treatments were arranged in a randomized complete block design with three replications per treatment.

Propagule density of *G. roseum* on the flowers was estimated immediately before any treatment was applied and at 2–3 d intervals thereafter. At each time of estimation, 10 flowers sampled randomly from each plot between 1700 and 1800 hours were placed in 200 mL sterile distilled water plus surfactant and propagules were estimated using the procedure described for the greenhouse studies. About 50 petals and 150 stamens of flowers were sampled from each plot as described above and placed on PCA to estimate incidence of *B. cinerea*. All ripe fruits (about 150–200 berries/replicate plot) harvested on 26 and 28 June were used for estimating gray mold fruit rot.

The field study was repeated once between 18 June and 31 July 1990 using strawberries grown in 1.2×1.2 m microplots located about 30 m from the strawberry planting. One hundred transplants of cv. Redcoat, each with a heavy crown, were stored over winter at -1°C , and planted in each microplot on 25 May 1990. The plants were supplied with 10-52-10 fertilizer and the soil was periodically irrigated without wetting the foliage. Treatments were arranged in a completely randomized design with four replicate microplots for each treatment. Spray treatments of spore suspensions were applied on 17, 24, and 29 June. Two half-sized hives with dispensers were positioned about 5 m apart among the microplots on 15 June, and inoculum was added to the dispensers daily from 17 June to 2 July. About 20–30% and 90–95% of the flowers had opened on the respective dates. Application of bee attractant, sampling of flowers, and disease estimations followed the same procedures as in the plot study. All ripe fruits (120–150 berries/plot) were harvested on 16 July for estimation of gray mold incidence.

Weather variables were monitored in the strawberry planting throughout the crop season and recorded using a data logger (model 21X, Campbell Scientific Inc., Logan UT 84321). Air temperature and r.h. were measured using shielded thermistors and r.h. probes, respectively (models 107 and 207, Campbell Scientific Inc.), positioned 30 cm above the soil. Rain was measured with a tipping-bucket gauge (Weathertronics Inc., Sacramento, CA 95841).

Data analyses. Statistical computations were performed using Statistical Analysis System software (SAS Institute, Cary, NC). Quantitative observations of *G. roseum* in controlled environment and the greenhouse, and of *B. cinerea* in the greenhouse were compared using a one-sided t-test. This test was used also to compare inoculum density of *G. roseum* in talc-cornmeal after various periods of cold storage. Observations in repetitions of experiments in controlled environment were tested for homogeneity of variance and pooled accordingly prior to analysis. Incidence data of *B. cinerea* on strawberry in the field were subjected to analysis of variance, and mean values were compared using the protected least significant difference (LSD) test (Snedecor & Cochran 1980). Field observations of *G. roseum* propagules on strawberry flowers in each replicate of each treatment at various times in the crop season were log-transformed and averaged; mean values were subjected to analysis of variance and separated by the protected LSD test. Data were detransformed for presentation. Linear correlation was used to examine relationships of estimated density of *G. roseum* and incidence of *B. cinerea* on strawberry flowers and fruits in the field. Linear correlation also was used to explore weather variables in relation to estimated density of *G. roseum* on flowers in field plots treated with the biocontrol agent.

Results

Inoculum production of *G. roseum*. Inoculum concentration was greater on corn meal than on oat bran, wheat bran, or poultry mash (Table 1). When sporulating cultures of *G. roseum* on PDA were dried and mixed with talc and corn meal, the formulation contained about 5×10^8 cfu/g. Concentration of viable propagules in the talc-corn meal formulation decreased progressively with time in storage at 4–5°C, except between the fourth and eighth weeks (Fig. 2). Numbers of cfu/g declined by 42.5% and 96.9% after 8 and 20 wk of refrigerated storage, respectively.

Inoculum acquisition by bees. Bees that crawled in corn meal and talc-corn meal (each with 10^8 cfu *G. roseum*/g) in petri dishes acquired 2.60–3.60 (mean 3.20×10^5 cfu/bee and 2.55–8.90 (mean 6.70) $\times$

Table 1. Inoculum production by *Gliocladium roseum* on various substrates

Substrate	cfu/g		cv. (%)
	Mean	Standard deviation	
Corn meal	8.2×10^7	3.0×10^6	3.2
Oat bran	2.3×10^7	5.5×10^6	23.9
Wheat bran	1.4×10^7	4.2×10^6	30.0
Poultry mash	1.5×10^6	2.7×10^5	18.0

10^5 cfu/bee, respectively. Bees emerging from the dispenser containing corn meal infested with 10^7 cfu *G. roseum*/g in the greenhouse carried $3.6\text{--}32.0$ (mean 6.30) $\times 10^4$ cfu of the fungus per bee. Those emerging from dispensers containing talc-corn meal infested with 5×10^8 cfu/g in the field each carried 9–180 (mean 57) $\times 10^4$ cfu. No *G. roseum* was detected on bees that crawled in noninfested substrates in petri dishes or dispensers.

Observations of contaminated bees. Macroscopically, infested substrate was visible as a whitish powder on external surfaces of bees that had crawled in the various substrates in petri dishes or dispensers (Fig. 3A). With the dissecting microscope, infested substrate was observed on external surfaces of almost all parts, including the head, mouthparts, antennae, thorax, wings, abdomen, and legs of bees that had crawled in corn meal and talc-corn meal formulations in petri dishes. Inoculum was especially dense on the bees' setae. Observations with the SEM indicated that numerous conidia of *G. roseum* and fragments of substrate adhered to femoral setae (Fig. 3 B-D) and to other parts of the legs. Conidia were often dense near the bases of the femoral setae (Fig. 3D). In some instances conidia appeared free from substrate and either separate or in clumps (Fig. 3C); in others, the spores adhered to substrate fragments (Fig. 3D).

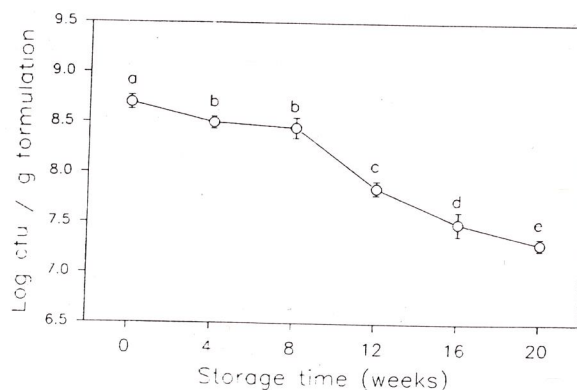


Figure 2. Viability of inoculum of *Gliocladium roseum* in a cornmeal-talc formulation with time in storage at 4–5°C.

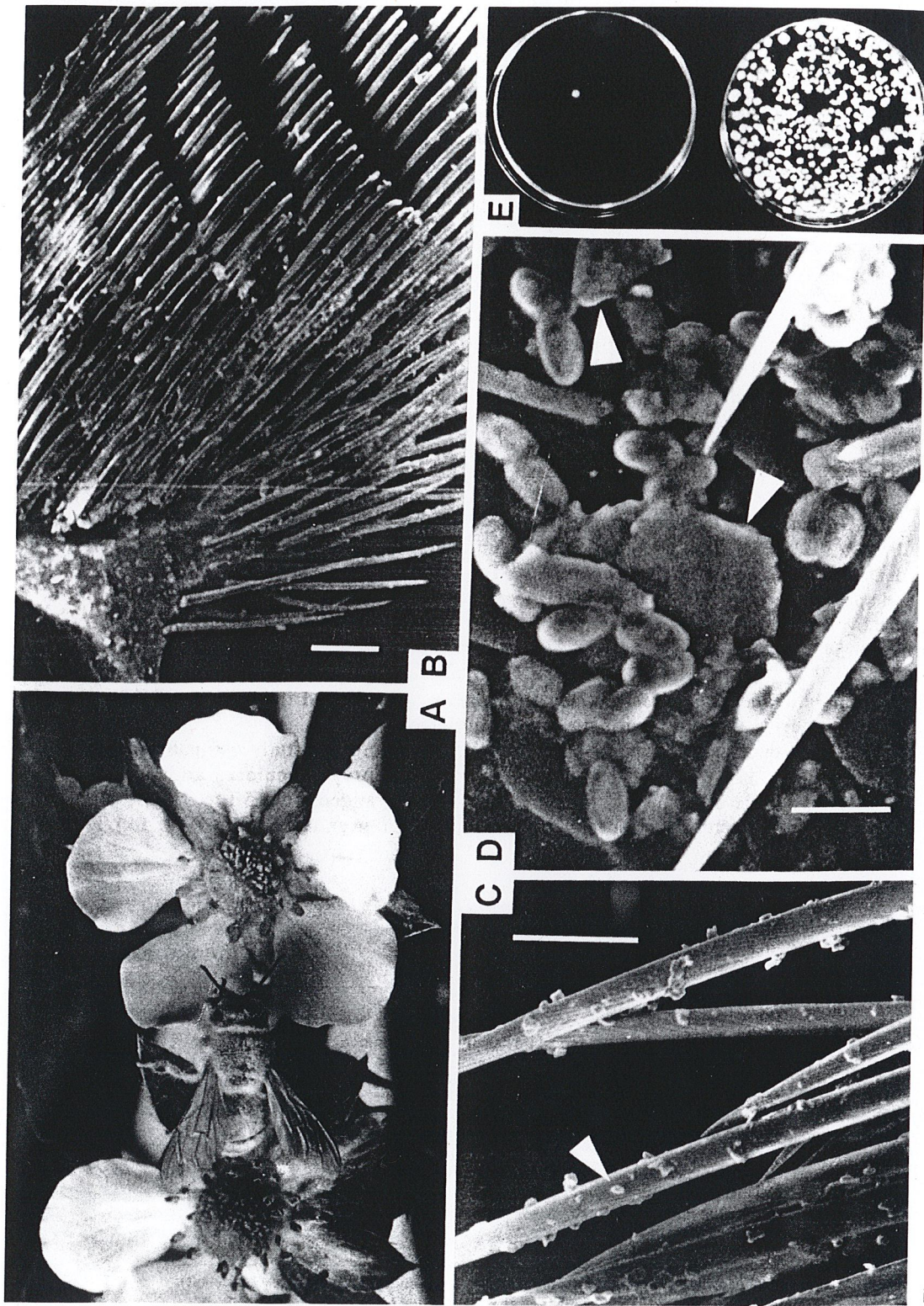


Figure 3. A) Honey bee contaminated with whitish inoculum of *Gliocladium roseum* on strawberry flowers; B-D) scanning-electron micrographs, of progressively increased magnification, showing femoral setae of a honey bee with adhering conidia and substrate of *G. roseum* (bars indicate 200, 70, and 15 µm in B, C, and D, respectively, and arrow heads point to a conidium in C and substrate particle in D); E) colonies produced on PD TSA by propagules recovered from untreated flowers (upper) and flowers infested with *G. roseum* by bee vectors (lower).

Studies in the greenhouse. The bees transferred large quantities of inoculum from the dispenser to the strawberry flowers (Table 2). About fourteen times more inoculum was transferred to flowers in the first trial, when bee activity was high, than in the second trial, when activity was low and mortality high. No *G. roseum* was detected on flowers when noninfested (check) substrates were used (Fig. 3E).

During foraging periods in the first trial, temperature ranged from 19 to 31°C (hourly mean 22.8°C) and r.h. from 43 to 71% (hourly mean 60%) when noninfested substrate was used; values when infested substrate was used were 19 to 31°C (22.4°C) and 33 to 67% r.h. (49% r.h.). Corresponding observations of the second trial were 20 to 29°C (22.6°C) and 38 to 60% r.h. (52% r.h.) for the noninfested substrate and 19 to 30°C (21.5°C) and 34 to 63% r.h. (51% r.h.) for the infested substrate.

Gliocladium roseum transferred to flowers by bees markedly suppressed sporulation potential of *B. cinerea* after the flowers were challenge-inoculated with the pathogen (Table 2). The biocontrol agent suppressed density of conidiophores of *B. cinerea* on the petals and sporulation incidence of the pathogen on the stamens by 47% and 54%, respectively, in the first trial, and by 54% and 47%, respectively, in the second trial.

Studies in the field. *Gliocladium roseum* was not detected on strawberry flowers prior to the inoculation studies. Low densities of the fungus were found during the studies, however, on flowers of untreated check plants (Table 3; means of 24 and 2 cfu/flower in the plots and microplots, respectively).

Estimated mean density of *G. roseum* on flowers visited by contaminated bees was about three times greater than that of spray-inoculated plants in the plot study, but not significantly different in the microplot study. Mean density of the biocontrol agent was higher on flowers inoculated by spraying or by bees in the plot study (4374 cfu/flower) than in the microplot study (747 cfu/flower).

In the plot study, about 10^4 cfu *G. roseum* were recovered per flower about 2 d after the strawberry plants were first inoculated, whether by spraying or by bees (Fig. 4A). Estimated density of the fungus in spray-inoculated and bee-inoculated treatments ranged from 300 to 15 000 and 1600 to 27 000 cfu/flower, respectively, during the study. In plants inoculated by spraying, density of *G. roseum* on the flowers oscillated, with increases associated with each application and followed by decreases. Rates of decline following successive spray applications (29–31 May, 2–6 June, and 8–12 June) were 36.0%, 30.3%, and 21.9% /day. Density of *G. roseum* unexpectedly increased between 12 and 14 June. Density of the fungus on flowers inoculated by bees was more stable than on spray-inoculated flowers. Density on bee-inoculated flowers was generally similar on 29 and 31 May, and 2, 4, 6, and 14 June, but lower on 8, 10, and 12 June. Density of *G. roseum* on untreated flowers ranged from 0 to 100 cfu/flower.

In the microplot study estimated density of *G. roseum* in spray-inoculated and bee-inoculated treatments ranged from 50 to 35 000 and 200 to 11 425 cfu/flower, respectively (Fig. 5A). Density of

Table 2. Estimated transfer of *Gliocladium roseum* from an inoculum dispenser to strawberry flowers by honey bees during two trials in the greenhouse, and effects of the transferred inoculum on estimated sporulation of *Botrytis cinerea* on the petals and stamens following challenge inoculation of the flowers with pathogen†

Trial	Type of substrate in inoculum dispenser	Inoculum density of <i>G. roseum</i> (cfu/flower)	Mean number of conidiophores of <i>B. cinerea</i> /cm ² petal	Sporulation incidence of <i>B. cinerea</i> on stamens (%)
1	Noninfested corn meal	0	137	58.1
	Infested corn meal‡	22 083δ**	73*	26.7**
2	Noninfested corn meal	0	114	28.6
	Infested corn meal‡	1542**	52**	15.1*

†Flowers were inoculated with *B. cinerea* (2×10^5 conidia/mL water plus surfactant) immediately after a 24-h period when bees were allowed to forage on the flowers.

‡About 10^7 cfu *G. roseum*/g corn meal.

δValues in a column for a given trial and followed by ** or * were significantly different at $P \leq 0.01$ and $P \leq 0.05$, respectively, according to a one-sided t test.

Table 3. Relationship of application method of *Gliocladium roseum* to strawberry plants and density of the fungus on the flowers in studies conducted in plots and microplots at Arkell in 1990

Application method	Mean cfu <i>G. roseum</i> /flower†	
	Plots	Microplots
None‡	24 c*	2 b
Inundative sprays‡§	2291 b	661 a
Bee vectors¶	6457 a	832 a

†Means of estimated values obtained for samples taken at 2-3 d intervals after inoculation treatments were initiated and for a 16- and 12-day study periods, respectively, in the plots and microplots. Individual observations were log transformed prior to pooling and subjected to an analysis of variance. The data were detransformed for presentation.

‡Plots were protected from bees with fibreglass screen.

§Suspension of *G. roseum* (10^7 conidia/mL water plus surfactant) was applied three times at intervals of 4-6 d and 5-6 d in the plots and microplots, respectively.

¶Bees were allowed to forage each day, and acquired inoculum from a dispenser containing a talc-corn meal formulation of *G. roseum* (5×10^8 cfu/g).

*Means in a column followed by the same letter are not significantly different ($P \leq 0.05$, protected LSD test).

the fungus in spray-inoculated plants oscillated in association with spray applications; rates of decline in periods following spray application (20-23 June and 25-29 June) were 32.8 and 24.6% /day. During 6 d following the spray inoculation of 23 June density of *G. roseum* declined to only 50 cfu/flower. Density of the fungus in bee-inoculated flowers did not fluctuate as widely as in spray-inoculated flowers. However, densities on 27 and 29 June and 2 July were lower than earlier in the study; the period of reduced density coincided with frequent foraging by the bees on nearby rapeseed (*Brassica napus* L.), which was then in flower.

Relationships of the monitored weather variables and density of *G. roseum* propagules on strawberry flowers were unclear (Fig. 4 A-D, Fig. 5 A-D). Daily temperature values did not vary widely throughout the studies (Fig. 4B, Fig. 5B). Daily humid periods that were short (3-7 h), intermediate (8-12 h), or long (13-24 h) did not appear to differentially affect patterns of propagule density of *G. roseum* on the flowers. Daily rainfall of 2-17 mm did not result in marked changes in estimated propagule density.

In the plot study, *G. roseum* often suppressed the incidence of *B. cinerea* as effectively, or more effectively, when applied by bees as when the fungus was applied as weekly sprays (Fig. 6). Both treatments markedly suppressed incidence of the pathogen on the petals when assessed on 29 and 31 May and 2, 6, and 8 June but not on 10 and 12 June. The bee-vectored fungus was effective also on 4 June. The two treatments each suppressed *B. cinerea*

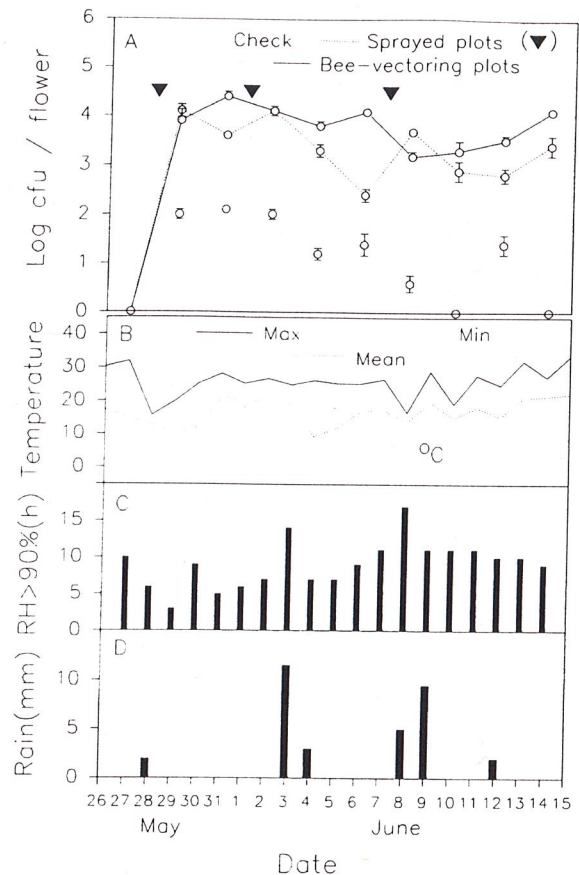


Figure 4. Variables monitored in strawberry plots in 1990: **A)** estimated inoculum density of *Gliocladium roseum* on strawberry flowers that were untreated, inoculated with spore suspensions of *G. roseum* applied as sprays, or inoculated with a talc-cornmeal formulation of *G. roseum* applied by honey bees; **B)** daily maximum, minimum, and mean hourly temperatures; **C)** daily periods when relative humidity exceeded 90%; and **D)** total daily rainfall.

on stamens sampled on 29 and 31 May and 2 June, and the spray application was effective also on 8 and 10 June. However, at other sampling times, incidence of the pathogen on stamens of treated plants and untreated plants did not differ significantly. Analysis of means for observations throughout the study indicated that *G. roseum* significantly suppressed incidence of *B. cinerea* on petals and stamens when applied as sprays or by bees, and that effectiveness of the methods did not differ significantly (Table 4). The two methods also did not differ significantly in effectiveness in suppressing *B. cinerea* on the fruits (Table 4).

In the microplot study, spray applications of *G. roseum* suppressed incidence of *B. cinerea* on the petals at all sampling times except 2 July, and on the

stamens (Fig. 7) at two sampling times (25 June and 2 July), but did not significantly suppress the pathogen on the fruits (Table 4). Application of *G. roseum* by bees failed to suppress the pathogen on petals, stamens, or fruits (Fig. 7, Table 4).

A significant negative correlation was found between the mean log cfu *G. roseum* per flower and incidence of *B. cinerea* on the petals ($R^2 = 0.93$, $n = 9$) and fruits ($R^2 = 0.68$, $n = 9$) in the plot study but not in the microplot study. Correlations between density of *G. roseum* on the flowers and incidence of *B. cinerea* on the stamens were not significant. Observations of flowers in all treatments were used in the analysis.

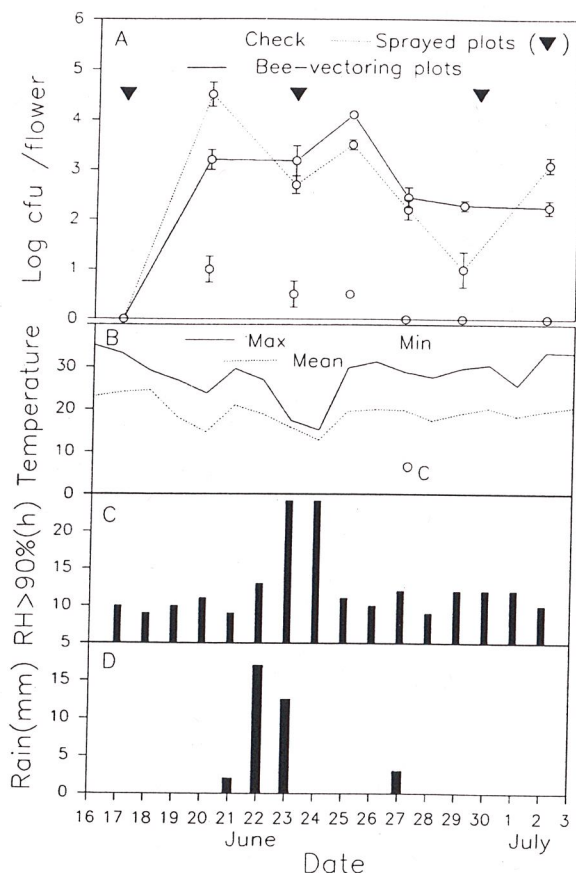


Figure 5. Variables monitored in microplots in 1990: A) estimated inoculum density of *Gliocladium roseum* on strawberry flowers that were untreated, inoculated with spore suspensions of *G. roseum* applied as sprays, or inoculated with a talc-cornmeal formulation of *G. roseum* applied by honey bees; B) daily maximum, minimum, and mean hourly temperatures; C) daily periods when relative humidity exceeded 90%; and D) total daily rainfall.

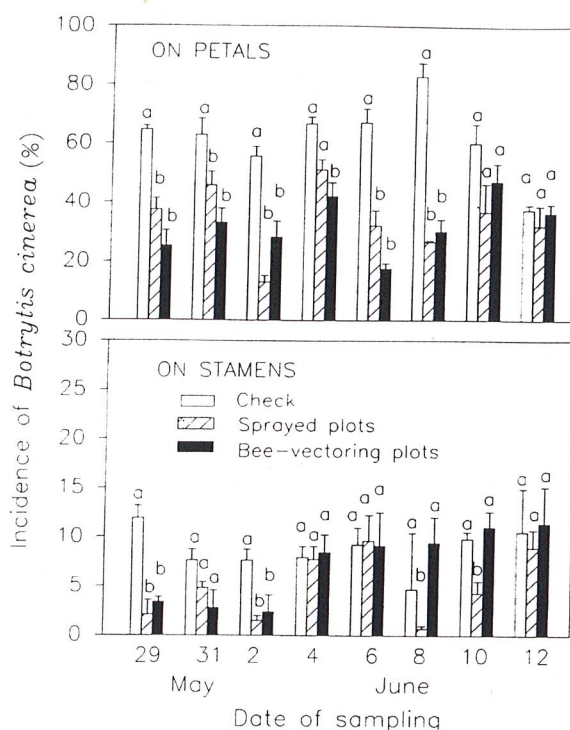


Figure 6. Incidence of *Botrytis cinerea* on petals and stamens of strawberry flowers that were untreated, or were treated with *Gliocladium roseum* applied as sprays or by honey bees, in field plots at Arkell in 1990. Data bars (each with a standard deviation bar) for petals or for stamens for a given sampling date assigned the same letter are not significantly different ($P \leq 0.05$, protected LSD test).

Discussion

The bees effectively transferred inoculum of *G. roseum* from the dispensers to strawberry flowers when conditions favored foraging activity. Each bee acquired about $3.6\text{--}32.0 \times 10^4$ cfu of *G. roseum* from the inoculum source containing 10^7 cfu/g in the greenhouse, and $9\text{--}180 \times 10^4$ cfu from the more concentrated source (5×10^8 cfu/g) used in the field. Thus, the bees acquired about 1 cfu for each 5000 cfu/g substrate. More than 2×10^4 and 6×10^3 cfu *G. roseum* were recovered per flower visited by bees when conditions were favorable for bee activity in the greenhouse and field, respectively. Density of the fungus on the flowers was about 10-fold lower, however, in the second study in the greenhouse when vigor of the bee colony was low, and in the microplot study in the field after nearby rapeseed began to bloom and to attract the bees. The *G. roseum* propagules that were recovered in low numbers from flowers of check strawberries not visited by bees probably were deposited chiefly by sedimentation from overflying bees; interplot dispersal of

Table 4. Relationships of application method of *Gliocladium roseum* to strawberry plants and incidence of *Botrytis cinerea* on the petals, stamens, and fruits of the strawberries in plots and microplots at Arkell in 1990

Application method	Incidence of <i>B. cinerea</i> (%)†					
	Plots			Microplots		
	Petals	Stamens	Fruits	Petals	Stamens	Fruits
None‡	53 a*	18 a	64 a	41 a	14 a	62 a
Inundative sprays‡§	37 b	12 b	44 b	27 b	11 a	52 a
Bee vectors¶	35 b	15 b	48 b	38 a	13 a	59 a

†Mean incidence for eight and four sampling times in the plots and microplots, respectively.

‡Plots were covered with fibreglass screen to prevent foraging by bees.

§Suspension of *G. roseum* (10^7 conidia/mL water plus surfactant) was applied three times at intervals of 4-6 d and 5-6 d in the plots and microplots, respectively.

¶Bees were allowed to forage daily and acquired inoculum from a dispenser containing a talc-corn meal formulation of *G. roseum* (5×10^8 cfu/g).

*Means in a column followed by the same letter are not significantly different ($P \leq 0.05$, protected LSD test).

propagules by rain, wind, or small insects was also possible. No *G. roseum* was detected on strawberry flowers before the treatments were first applied or in the same plot area in an earlier study (McLean 1988).

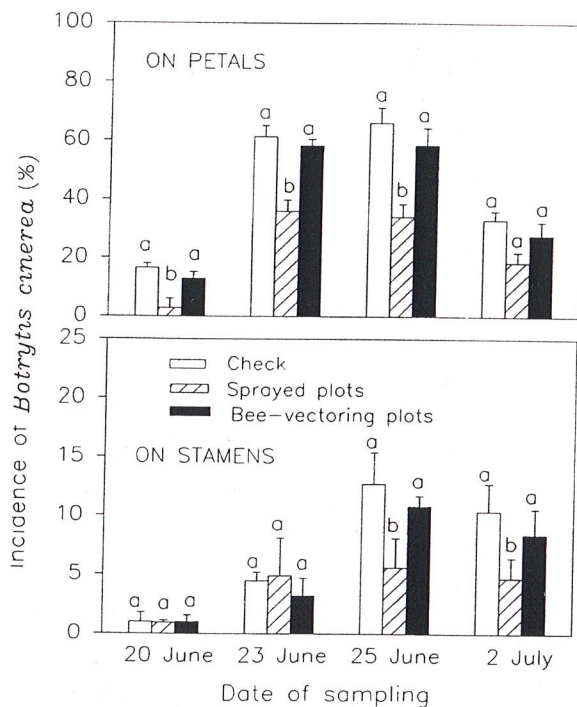


Figure 7. Incidence of *Botrytis cinerea* on petals and stamens of strawberry flowers that were untreated, or were treated with *Gliocladium roseum* applied as sprays or by honey bees, in microplots at Arkell in 1990. Data bars (each with a standard deviation bar) for petals or for stamens for a given sampling date assigned the same letter are not significantly different ($P \leq 0.05$, protected LSD test).

Density of *G. roseum* propagules on flowers in the field studies usually was as high or higher when applied by bees as in spray-treated strawberries. Propagule density generally remained high and stable on bee-inoculated flowers but oscillated markedly in the spray treatment. Frequent visits of bees to the flowers each day likely ensured deposition of propagules on freshly-opened flowers and maintenance of high densities of propagules on previously-visited flowers. In contrast, mean densities of the fungus on sprayed flowers declined sharply during several days following each application. During these periods, viability of propagules on treated flowers may have fallen, and untreated flowers that opened after spraying likely lacked inoculum. Relative stability of propagule density on flowers inoculated by bees and sharp declines on spray-inoculated flowers were observed in a range of weather conditions. In the microplot study, however, propagule density was consistently low when bees preferentially visited flowering rapeseed. Although the two methods of application differed in mechanism and in inoculum concentration used, the observations underscored the effectiveness of bees as an alternative to sprays for applying the biocontrol agent.

Inoculum of *G. roseum* applied by bees generally suppressed incidence of *B. cinerea* on petals, stamens, and fruits as effectively as spray applications. In some instances, applications by bees suppressed *B. cinerea* on petals more effectively than did spray applications, and the reverse was sometimes found for petals or for stamens (Figs. 6, 7). Incidence of the pathogen on petals and stamens in the microplots was not suppressed by the bee treatment after the bees began preferential visits to

the rapeseed. In the microplots, *G. roseum* failed to suppress incidence of *B. cinerea* on the fruits when applied by bees or as sprays, thus factors in addition to a paucity of bee visits may have interfered with biocontrol. Growth cracks and associated increase in nutrient exudation possibly developed in the unusually large and succulent fruits produced by the heavy-crowned plants grown in the microplots; cracks and nutrients could have favored direct infection of the fruits, before or after harvest, by conidia of *B. cinerea* (Jarvis 1962, David & Dennis 1979), against which a biocontrol agent applied to the flowers would not likely be effective. In the plot study, *G. roseum* applied by bees or as sprays suppressed incidence of *B. cinerea* in the fruits only moderately, and less than in earlier studies (Peng & Sutton 1991a). However, density of airborne conidia from untreated strawberry plants in an area immediately surrounding the plots was likely high, and the fruits were evaluated for incidence of the pathogen only after prolonged storage (15 d) in moist conditions. Although density of *G. roseum* propagules on strawberry flowers was often higher and more stable following application by bees compared to spray applications, *B. cinerea* was suppressed only just as effectively by the bee treatments.

Setae on the bees and stickiness of *G. roseum* conidia were probably important factors contributing to the abundant transmission of the biocontrol agent. From the microscopic observations, inoculum was especially copious on portions of infested bees with abundant setae, and numerous spores adhered directly to the femoral setae. Collectively, the setae present a large surface area to which inoculum may adhere; however adhesion to other external surfaces of bees may also be important for inoculum transmission. Stickiness of *G. roseum* conidia (Barron 1968) may facilitate adhesion to bees. Transfer of inoculum from the bee to various parts of the strawberry flower is not well understood, and probably depends in part on activities of the bee while on the flower (Free 1968b), and on the surface characteristics of the flowers.

The quantitative relationships of inoculum transfer by bees and the effects of transferred inoculum on gray mold indicated that a high concentration of *G. roseum* may be needed at the source in order to maximize biocontrol. In the bee treatments, moderate suppression of gray mold was observed when several thousand propagules of *G. roseum* were recovered per flower, but biocontrol was ineffective when only a few hundred propagules were present. A high concentration of *G. roseum* at the source may help to counter the impact of variables that suppress bee activity and inoculum density on strawberry flowers.

Increased concentration of the inoculum source from 1×10^7 to 5×10^8 cfu/g generally brought about proportionate increase in inoculum acquisition by the bees, and greater increases in inoculum loads carried by bees may be possible through higher concentration of the inoculum or by modifying the particle size, stickiness, or other properties of the substrate. Maintenance of high concentrations of viable inoculum appears feasible based on the observation that 58% of propagule viability of *G. roseum* in the talc-corn meal formulation remained after 8 wk at 5°C; improved maintenance of viability may be possible at lower temperature, by use of controlled water potential, or by other means (Van den Berg & Lentz 1968, Kenny & Couch 1981, Papavizas et al. 1984, Harman et al. 1990).

It is concluded that honey bees are effective vehicles for delivering inoculum of *G. roseum* to strawberry flowers for biocontrol of gray mold. Bees maintained high populations of the biocontrol agent on strawberry flowers except when visits to flowers were infrequent. Cool temperatures, wind, and rain reduced foraging by bees on flowers, but only for brief periods, and no marked changes in density of *G. roseum* were observed on the flowers in relation to these conditions. However, infrequent foraging by honeybees on strawberries during several days, as observed while the rapeseed bloomed, may result in insufficient maintenance of *G. roseum* on the flowers for effective biocontrol of gray mold. From this observation, bees may also be ineffective for delivering inoculum should weather conditions discourage foraging for prolonged periods. Spray applications of *G. roseum* may be needed during periods unfavorable for foraging, especially if conditions also strongly favor infection by *B. cinerea* (Bulger et al. 1987, Ellis & Madden 1991). Improved attractants for bees applied to strawberries may help to avoid the normally preferential visits to flowers of various other kinds of plants.

Insofar as the authors are aware, this is the first detailed report of the use of bees to deliver a biocontrol agent to plants. Preliminary reports of this work (Peng & Sutton 1991b) and of the use of bees to deliver bacteria to pear and apple flowers for biocontrol of fireblight (Thomson et al. 1990) are available. The method holds promise for direct application of various microorganisms to strawberry and many other hosts for the biocontrol of flower-infecting pathogens.

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